

# Indian Institute of Technology, Madras - Centre for Continuing Education

## Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

|                                         |                                       |
|-----------------------------------------|---------------------------------------|
| Question Paper Name :                   | IIT M QUALIFIER EXAM QPQ2 26 Feb 2023 |
| Subject Name :                          | 2023 Feb: IIT M QUALIFIER EXAM QPQ2   |
| Creation Date :                         | 2023-02-22 16:43:20                   |
| Duration :                              | 240                                   |
| Total Marks :                           | 190                                   |
| Display Marks:                          | Yes                                   |
| Share Answer Key With Delivery Engine : | Yes                                   |
| Actual Answer Key :                     | Yes                                   |
| Calculator :                            | Scientific                            |
| Magnifying Glass Required? :            | No                                    |
| Ruler Required? :                       | No                                    |
| Eraser Required? :                      | No                                    |
| Scratch Pad Required? :                 | No                                    |
| Rough Sketch/Notepad Required? :        | No                                    |
| Protractor Required? :                  | No                                    |
| Show Watermark on Console? :            | Yes                                   |
| Highlighter :                           | No                                    |
| Auto Save on Console?                   | Yes                                   |
| Change Font Color :                     | No                                    |
| Change Background Color :               | No                                    |

|                     |    |
|---------------------|----|
| Change Theme :      | No |
| Help Button :       | No |
| Show Reports :      | No |
| Show Progress Bar : | No |

Group I

|                                                 |             |
|-------------------------------------------------|-------------|
| Group Number :                                  | 1           |
| Group Id :                                      | 64065311130 |
| Group Maximum Duration :                        | 0           |
| Group Minimum Duration :                        | 90          |
| Show Attended Group? :                          | No          |
| Edit Attended Group? :                          | No          |
| Break time :                                    | 0           |
| Group Marks :                                   | 190         |
| Is this Group for Examiner? :                   | No          |
| Examiner permission :                           | Cant View   |
| Show Progress Bar? :                            | No          |
| Revisit allowed for group Instructions? :       | Yes         |
| Maximum Instruction Time :                      | 0           |
| Minimum Instruction Time :                      | 0           |
| Group Time In :                                 | Minutes     |
| Navigate To Group Summary From Last Question? : | No          |
| Disable Submit Button During Assessment? :      | No          |
| Section Selection Time? :                       | 0           |
| No of Optional sections to be attempted :       | 0           |

|                                                                     |             |
|---------------------------------------------------------------------|-------------|
| <b>Section Id :</b>                                                 | 64065330299 |
| <b>Section Number :</b>                                             | 1           |
| <b>Section type :</b>                                               | Online      |
| <b>Mandatory or Optional :</b>                                      | Mandatory   |
| <b>Number of Questions :</b>                                        | 12          |
| <b>Number of Questions to be attempted :</b>                        | 12          |
| <b>Section Marks :</b>                                              | 50          |
| <b>Display Number Panel :</b>                                       | Yes         |
| <b>Group All Questions :</b>                                        | No          |
| <b>Enable Mark as Answered Mark for Review and Clear Response :</b> | Yes         |
| <b>Maximum Instruction Time :</b>                                   | 0           |
| <b>Sub-Section Number :</b>                                         | 1           |
| <b>Sub-Section Id :</b>                                             | 64065367461 |
| <b>Question Shuffling Allowed :</b>                                 | No          |
| <b>Is Section Default? :</b>                                        | null        |

**Question Number : 1 Question Id : 640653469242 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 0**

Question Label : Multiple Choice Question

**THIS IS QUESTION PAPER FOR THE SUBJECT "QUALIFIER LEVEL : MATHEMATICS FOR DATA SCIENCE I"**

**MANDATORILY YOU HAVE TO ATTEND ALL THE SECTIONS**

**Options :**

6406531559606. ✓ YES

6406531559607. ✗ NO

**Question Number : 2 Question Id : 640653469243 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 0**

Question Label : Multiple Choice Question

### Instructions:

- There are some questions which have functions with discrete valued domains (such as day, month, year etc). For simplicity, we treat them as continuous functions.
- For NAT type question, enter only one right answer even if you get multiple answers for that particular question.
- Notations:
  - $\mathbb{R}$  = Set of real numbers
  - $\mathbb{Q}$  = Set of rational numbers
  - $\mathbb{Z}$  = Set of integers
  - $\mathbb{N}$  = Set of natural numbers
- The set of natural numbers includes 0.

### Options :

6406531559608. ✓ Useful Data has been mentioned above

6406531559609. ✗ This data attachment is just for a reference & not for an evaluation.

|                                     |             |
|-------------------------------------|-------------|
| <b>Sub-Section Number :</b>         | 2           |
| <b>Sub-Section Id :</b>             | 64065367462 |
| <b>Question Shuffling Allowed :</b> | Yes         |
| <b>Is Section Default? :</b>        | null        |

**Question Number : 3 Question Id : 640653469244 Question Type : SA Calculator : None**

**Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4**

Question Label : Short Answer Question

The Cartesian product  $A \times A$  has 9 elements. Two of the elements of the Cartesian product are (2, 0) and (0, 3). Find the sum of all the elements in set A.

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

5

**Sub-Section Number :** 3

**Sub-Section Id :** 64065367463

**Question Shuffling Allowed :** Yes

**Is Section Default? :** null

**Question Number : 4 Question Id : 640653469245 Question Type : SA Calculator : None**

**Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5**

Question Label : Short Answer Question

In a survey among 260 students in Nilgiri house of IITM BSc degree, the following data were found:

- 100 students have a Hotstar subscription
- 110 students have Netflix subscription
- 120 students have Amazon Prime membership.
- 30 students have both Hotstar and Netflix subscriptions whereas 30 students have both Hotstar and Amazon Prime and 40 students subscribe to both Netflix and Amazon Prime.

Assuming that all students have at least one OTT subscription, determine how many students have memberships to all 3 OTT: Hotstar, Netflix and Amazon Prime?

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

30

Question Number : 5 Question Id : 640653469247 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

Assume that a ball was thrown from the point  $(0, 5)$  on the  $XY$ -plane as shown in the Figure 1. The ball reaches a maximum height of 7 meters and it returns to a height of 5 m after 2 seconds. Let  $h(t) = at^2 + bt + c$  be the quadratic function that represents the height (in meters) of the ball after  $t$  seconds, where  $a, b, c \in \mathbb{Z} \setminus \{0\}$ .

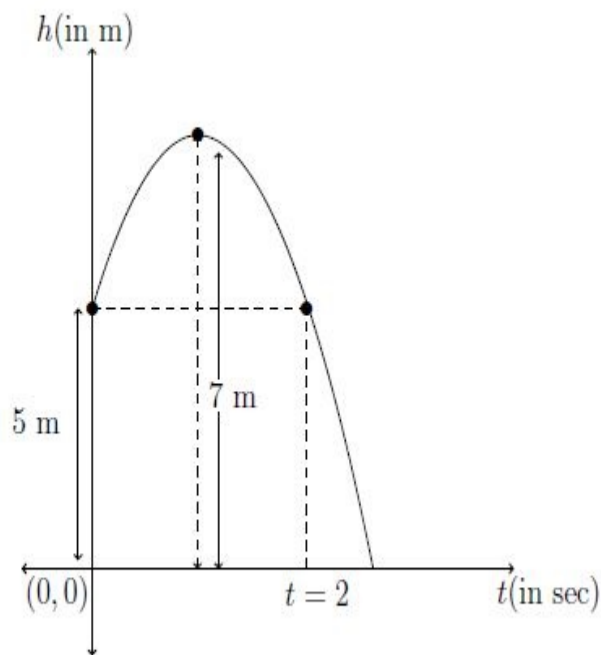


Figure 1

Find the values of  $a + b + c$ .

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Question Number : 6 Question Id : 640653469250 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

You have been closely monitoring your bike’s mileage recently. Here is a table showing two rows representing the amount paid for fuel(in ₹) and the corresponding mileage (in Km). Consider  $y$  as the amount paid and  $x$  as the corresponding mileage in Km. You have noted down the distance traveled each time when the fuel meter falls back to a fixed reference mark and predicted that the equation of the best fit line is  $y = 5x - 22$ . What will be the value of SSE w.r.t the best fit line?

|                    |    |    |    |     |    |
|--------------------|----|----|----|-----|----|
| Amount paid (in ₹) | 80 | 50 | 60 | 100 | 49 |
| Distance (in Km)   | 20 | 15 | 16 | 25  | 14 |

Table: 1

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

27

Sub-Section Number : 4

Sub-Section Id : 64065367464

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 7 Question Id : 640653469246 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

Suppose  $A = \{a, b, c, d\}$  and  $B = \{p, q, r, s\}$  are two sets. Consider the following relations from  $A$  to  $B$ .

- $R_1 = \{(a, p), (c, r), (d, q)\}$
- $R_2 = \{(a, s), (b, s), (c, p), (d, r)\}$
- $R_3 = \{(a, p), (b, r), (b, s), (d, q)\}$
- $R_4 = \{(a, r), (b, p), (c, q), (d, s)\}$

Which of the following statements are correct?

**Options :**

6406531559612. ✖  $R_2, R_3$ , and  $R_4$  are functions.

6406531559613. ✔  $R_2$  and  $R_4$  are functions.

6406531559614. ✖  $R_2$  is an injective function.

6406531559615. ✔  $R_4$  is a bijective function.

**Question Number : 8 Question Id : 640653469248 Question Type : MSQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5 Selectable Option : 0**

Question Label : Multiple Select Question

Which of the following functions may represent the graph given in Figure 2?

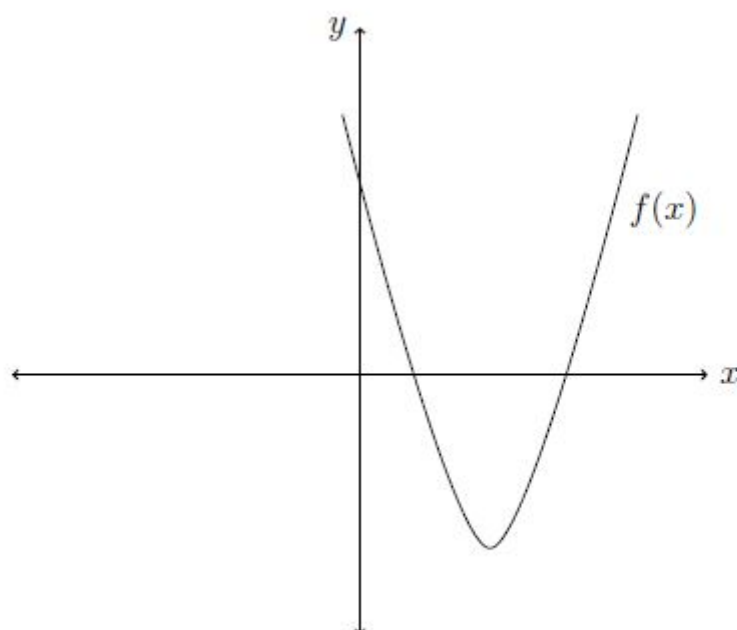


Figure: 2



**Options :**

6406531559617. ✓  $f(x) = x^2 - 8x + 12$

6406531559618. ✗  $f(x) = x^2 + 10x - 21$

6406531559619. ✗  $f(x) = 2x^2 + 8x + 4$

6406531559620. ✓  $f(x) = x^2 - 6x + 5$

**Question Number : 9 Question Id : 640653469249 Question Type : MSQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5 Selectable Option : 0**

Question Label : Multiple Select Question

Ankit is located at (3, 3). He called Ajay to ask his location. Ajay describes the path he had taken from home (located at the origin) as: "I walked three units towards East and then nine units towards North. And I repeated the same pattern thrice." Now Ankit wants a direct path to reach Ajay, then choose the correct options.(Note that North represents the direction along the positive y-axis.)

**Options :**

6406531559621. ✓ Ankit should follow  $4x - y - 9 = 0$ .

6406531559622. ✓ Ankit will have to walk a distance of  $6\sqrt{17}$  units

6406531559623. ✓ Ajay has walked a distance of 36 units from his home.

6406531559624. ✗ Ajay has walked a distance of  $9\sqrt{10}$  units from his home.

**Question Number : 10 Question Id : 640653469252 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5 Selectable Option : 0**

Question Label : Multiple Select Question

Rubika launches her new company in the year 2010, which makes a yearly profit in lakhs as the polynomial function  $p(x) = 0.1x^2(x-1)(x-2)(x-3)^2(x-4)(x-9)$  for the first 12 years since the launch, where  $x$  is the number of years since 2010 (i.e.,  $x = 0$  denotes the year 2010,  $x = 1$  denotes the year 2011, and so on). Let the loss be represented as  $-ve$  of profit. Which of the following options are correct?

**Options :**

6406531559630. ✓ Including the year in which the company was launched, it neither made a profit nor a loss six times in 12 years.

6406531559631. ✗ The company made profit in years  $x \in \{7, 8, 10, 11\}$ .

6406531559632. ✓ The company made loss ( $-ve$  profit) in years  $x \in \{5, 6, 7, 8\}$ .

6406531559633. ✓ In the year 2022 (i.e.,  $x = 12$ ) the company made profit.

**Question Number : 11 Question Id : 640653469253 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5 Selectable Option : 0**

Question Label : Multiple Select Question

Figure 4 shows the graph of a polynomial  $p(x)$ . Choose the set of correct option(s).

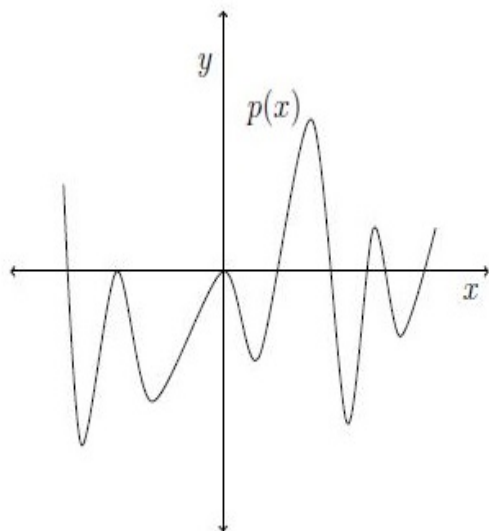


Figure 4

**Options :**

6406531559634. ✖ The degree of  $p(x)$  is at least 9.

6406531559635. ✔  $p(x)$  represent an even degree polynomial.

6406531559636. ✔ Total number of turning point of  $p(x)$  are 9.

6406531559637. ✔ Multiplicities of zero and one of the negative roots could be the same.

**Sub-Section Number :** 5

**Sub-Section Id :** 64065367465

**Question Shuffling Allowed :** Yes

**Is Section Default? :** null

**Question Number : 12 Question Id : 640653469251 Question Type : MSQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 6 Selectable Option : 0**

**Question Label : Multiple Select Question**

Figure: 3 shows the curves represented by polynomials  $f(x)$ ,  $g(x)$ , and  $h(x)$  of degrees 4, 4, and 2 respectively, on the  $XY$  plane. Let  $f(x) - g(x) = ax(x - 2)(x - 5)(x - 11)$ ,  $a \neq 0$ . If  $b$  is a negative constant, then choose the most possible expression for  $h(x)$  and other correct statements among the given options. (Note that the figure is not according to scale .)

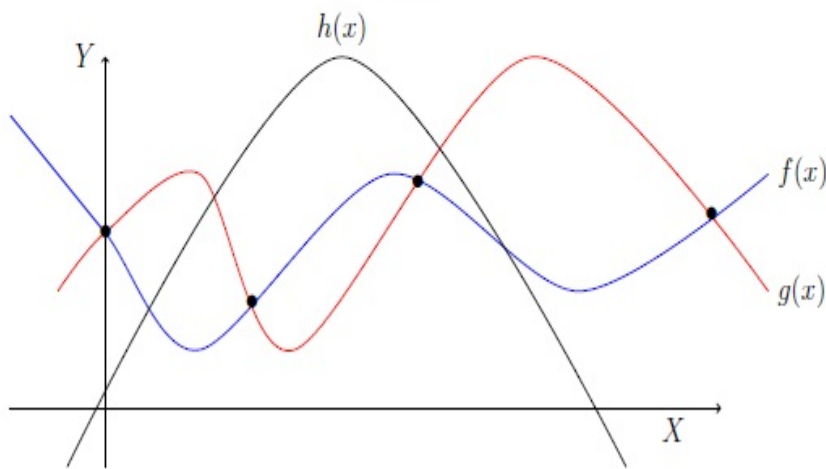


Figure: 3

Options :

6406531559626. ✖  $h(x) = b(x^2 + 8x - 7)$

6406531559627. ✖  $f(x) = g(x)$  at  $x = 0, -2, -5, -11$

6406531559628. ✔  $h(x) = b(x^2 - 6x - 7)$

6406531559629. ✔  $f(x) = g(x)$  at  $x = 0, 2, 5, 11$

## English 1

|                                       |             |
|---------------------------------------|-------------|
| Section Id :                          | 64065330300 |
| Section Number :                      | 2           |
| Section type :                        | Online      |
| Mandatory or Optional :               | Mandatory   |
| Number of Questions :                 | 22          |
| Number of Questions to be attempted : | 22          |

|                                                              |             |
|--------------------------------------------------------------|-------------|
| Section Marks :                                              | 50          |
| Display Number Panel :                                       | Yes         |
| Group All Questions :                                        | No          |
| Enable Mark as Answered Mark for Review and Clear Response : | Yes         |
| Maximum Instruction Time :                                   | 0           |
| Sub-Section Number :                                         | 1           |
| Sub-Section Id :                                             | 64065367466 |
| Question Shuffling Allowed :                                 | No          |
| Is Section Default? :                                        | null        |

Question Number : 13 Question Id : 640653469254 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "QUALIFIER LEVEL : ENGLISH I"  
MANDATORILY YOU HAVE TO ATTEND ALL THE SECTIONS

Options :

6406531559638. ✓ YES

6406531559639. ✗ NO

|                              |             |
|------------------------------|-------------|
| Sub-Section Number :         | 2           |
| Sub-Section Id :             | 64065367467 |
| Question Shuffling Allowed : | No          |
| Is Section Default? :        | null        |

Question Id : 640653469255 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (14 to 18)

Question Label : Comprehension

**Read the following passage and answer the given subquestions.**

My vision for India in the 100th year of its Independence is that of a global innovation leader focused on equitable and inclusive economic growth where everyone prospers in a secure environment with equal access to healthcare, education and sustainable livelihoods. It is by investing in breakthrough ideas and embracing entrepreneurship as an economic model of growth that India will be able to unleash the power of innovation to ensure a better life for all its citizens. Research and innovation, fuelled by technology, will catalyse the kind of non-linear growth that will make the country one of the world's top three economies and bring it closer to developed nation status by 2047.

The government has identified research, innovation and technology as the key drivers of its VisionIndia@2047 and is making significant investments in these areas. Going forward, it must enable and support innovative start-ups and businesses that think locally but have the potential to make an enormous global impact. By encouraging technopreneurs to grow from small and medium enterprises (SMEs) to large industrial-scale operations, India will be able to create a compelling opportunity to take innovative ideas to global markets. We should aim to be among the top 20 countries in the Global Innovation Index by 2047. Digital and data-backed innovation combined with ubiquitous and affordable internet will help India build a robust digital economy of the future.

Given India's value advantage and scientific excellence, the country can innovate and drive the technology frontier outward to boost productivity and efficiency in the sphere of healthcare, thus delivering affordability and accessibility, enabling preventive health approaches and driving value across the complete spectrum of the healthcare ecosystem. Technology-led innovations will help India leapfrog the traditional linear model and take healthcare to the next level. It will help us build a national healthcare system, which is accessible, organised, accountable, affordable and, thus, resilient.

The groundwork for a digital transformation of healthcare in India has already been laid, thanks to the reforms being implemented by the Centre to make healthcare-related regulations more flexible. At the same time, the pandemic has forced changes in consumer behaviour, persuaded doctors to become tech-savvy and prompted the healthcare industry to invest in user-friendly digital solutions. These developments have created a fertile environment for the emergence of

low-cost, technology-led and scalable innovations anchored in affordability and accessibility. To reap the full benefits of this digital disruption of healthcare, India will need to coordinate policymaking, funding and implementation. It will also need to put in place appropriate regulatory frameworks that foster patient trust and privacy, as well as enable faster adoption of emerging technological innovations.

### Sub questions

**Question Number : 14 Question Id : 640653469256 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

An appropriate replacement for the word *ubiquitous* is \_\_\_\_\_.

**Options :**

6406531559640. ✓ Universal

6406531559641. ✗ Unrestrained

6406531559642. ✗ Autonomous

6406531559643. ✗ Limited

**Question Number : 15 Question Id : 640653469257 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Choose an appropriate replacement for the word *leapfrog*.

**Options :**

6406531559644. ✓ Catapult

6406531559645. ✗ Catalyst

6406531559646. ✗ Trailing

6406531559647. ✗ Increase

**Question Number : 16 Question Id : 640653469258 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

The pandemic has prompted \_\_\_\_\_.

**Options :**

- 6406531559648. ✖ Doctors to be tech-savvy
- 6406531559649. ✖ Incentivised industrial investments in user-friendly digital solutions
- 6406531559650. ✖ Low cost health solutions
- 6406531559651. ✔ All of these

**Question Number : 17 Question Id : 640653469259 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

How could India utilise the technological advancements in healthcare?

**Options :**

- 6406531559652. ✔ Coordinating policy making, funding, and implementation
- 6406531559653. ✖ Expanding the benefits of technology
- 6406531559654. ✖ Converting health sector into digitally powered one
- 6406531559655. ✖ All of these

**Question Number : 18 Question Id : 640653469260 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Patient trust and privacy can be inculcated by \_\_\_\_\_.

**Options :**



6406531559656. ✓ Adequate regulatory frameworks

6406531559657. ✗ Ensuring digital access

6406531559658. ✗ Implementing technological solutions

6406531559659. ✗ Improving the access of healthcare services

**Question Id : 640653469261 Question Type : COMPREHENSION Sub Question Shuffling  
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix  
Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0  
Question Numbers : (19 to 23)**

Question Label : Comprehension

**Listen to the audio sample and answer the given subquestions:**



885\_640653\_0\_1984128\_hs1001qz1ane2q6mq.mp3

**Sub questions**

**Question Number : 19 Question Id : 640653469262 Question Type : MCQ Is Question  
Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction  
Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Who was the speaker for Natalie Portman's class day?

**Options :**

6406531559660. ✓ Will Ferrell

6406531559661. ✗ Tom Hanks

6406531559662. ✗ Will Smith

6406531559663. ✗ Al Pacino

**Question Number : 20 Question Id : 640653469263 Question Type : MCQ Is Question  
Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction  
Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

When invited to give a Harvard commencement speech, what response did Natalie Portman give as an email?

**Options :**

6406531559664. ✔ *"Wow, this is so nice. I'm gonna need some funny ghost writers. Any ideas?"*

6406531559665. ✖ *"Wow, this is so amazing. I'm honoured to be invited"*

6406531559666. ✖ *"Wow, this is great. I will be there"*

6406531559667. ✖ *"Wow, this is so cool"*

**Question Number : 21 Question Id : 640653469264 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

When did Natalie Portman enrol at Harvard?

**Options :**

6406531559668. ✖ 1989

6406531559669. ✔ 1999

6406531559670. ✖ 2001

6406531559671. ✖ 2002

**Question Number : 22 Question Id : 640653469265 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

As per the audio, Natalie Portman's soon-to-be son was playing \_\_\_\_\_.

**Options :**

6406531559672. ✖ Basketball

6406531559673. ✔ Arcade games

6406531559674. ✖ Tennis

6406531559675. ✖ Badminton

**Question Number : 23 Question Id : 640653469266 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Natalie Portman won an Oscar.

**Options :**

6406531559676. ✔ True

6406531559677. ✖ False

**Sub-Section Number :** 3

**Sub-Section Id :** 64065367468

**Question Shuffling Allowed :** No

**Is Section Default? :** null

**Question Id : 640653469267 Question Type : COMPREHENSION Sub Question Shuffling**

**Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix**

**Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Question Numbers : (24 to 28)**

Question Label : Comprehension

**Read the following telephonic conversation and fill in the blank with appropriate responses for the given subquestions:**

Matt: Hello. Good morning. I am Matt. Am I speaking to Mr. John?

Susan: Hi. Good morning. I am Susan.

Matt: (i) \_\_\_\_\_

Susan: Sure.

Matt: (ii) \_\_\_\_\_

Susan: (iii)\_\_\_\_\_. I couldn't find John around.

Matt: Okay, (iv)\_\_\_\_\_

Susan: Sure. (v) \_\_\_\_\_

Matt: Thanks.

### Sub questions

**Question Number : 24 Question Id : 640653469268 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete blank (i) with an appropriate response.

**Options :**

6406531559678. ✖ I want to talk to John

6406531559679. ✔ Could you please connect me to John

6406531559680. ✖ How are you?

6406531559681. ✖ How is your life?

**Question Number : 25 Question Id : 640653469269 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete blank (ii) with an appropriate response.

**Options :**

6406531559682. ✔ Thank you

6406531559683. ✖ Okay

6406531559684. ✖ Good

6406531559685. ✖ Sure

**Question Number : 26 Question Id : 640653469270 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete blank (iii) with an appropriate response.

**Options :**

6406531559686. ✖ I can't

6406531559687. ✖ He is not here

6406531559688. ✖ Please wait

6406531559689. ✔ Sorry

**Question Number : 27 Question Id : 640653469271 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete blank (iv) with an appropriate response.

**Options :**

6406531559690. ✖ Thank you

6406531559691. ✔ Could I leave him a message

6406531559692. ✖ I want to talk to him

6406531559693. ✖ Could you please call him

**Question Number : 28 Question Id : 640653469272 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete blank (v) with an appropriate response.

**Options :**

6406531559694. ✔ Let me jot it down

6406531559695. ✖ I will inform him

6406531559696. ✖ Call back after sometime

6406531559697. ✖ Thanks

**Question Id : 640653469278 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Question Numbers : (29 to 33)**

Question Label : Comprehension

**Match Column A with suitable options in Column B. (Hint: Word collocation)**

| A            | B              |
|--------------|----------------|
| 1. Land      | a) Forecast    |
| 2. Moral     | b) Decision    |
| 3. Final     | c) Acquisition |
| 4. Weather   | d) Blow        |
| 5. Unanimous | e) Decay       |

Based on the above data, answer the given subquestions.

**Sub questions**

**Question Number : 29 Question Id : 640653469279 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Land \_\_\_\_

**Options :**

6406531559713. ✖ Forecast

6406531559714. ✖ Decision

6406531559715. ✔ Acquisition

6406531559716. ✖ Blow

6406531559717. ✖ Decay

**Question Number : 30 Question Id : 640653469280 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Moral \_\_\_\_

**Options :**

6406531559718. ✖ Forecast

6406531559719. ✖ Decision

6406531559720. ✖ Acquisition

6406531559721. ✖ Blow

6406531559722. ✔ Decay

**Question Number : 31 Question Id : 640653469281 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Final \_\_\_\_

**Options :**

6406531559723. ✖ Forecast

6406531559724. ✖ Decision

6406531559725. ✖ Acquisition

6406531559726. ✔ Blow

6406531559727. ✖ Decay

**Question Number : 32 Question Id : 640653469282 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Weather \_\_\_\_

**Options :**

6406531559728.  Forecast

6406531559729.  Decision

6406531559730.  Acquisition

6406531559731.  Blow

6406531559732.  Decay

**Question Number : 33 Question Id : 640653469283 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Unanimous \_\_\_\_

**Options :**

6406531559733.  Forecast

6406531559734.  Decision

6406531559735.  Acquisition

6406531559736.  Blow

6406531559737.  Decay

**Sub-Section Number :** 4

**Sub-Section Id :** 64065367469

**Question Shuffling Allowed :** No

**Is Section Default? :** null

**Question Id : 640653469273 Question Type : COMPREHENSION Sub Question Shuffling**

**Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix**



**Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Question Numbers : (34 to 37)**

Question Label : Comprehension

**Context: A new development project is going to come up in your area that involves a lot of displacement of people. Different stakeholders with the project are stating their opinion on the various impacts, and the pros and cons of the project. Answer the given subquestions.**

**Sub questions**

**Question Number : 34 Question Id : 640653469274 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

A social activist is stating her opinion on this. The perspective she would choose is:

**Options :**

6406531559698. ✖ Making the masterplan larger

6406531559699. ✖ Giving refreshments to the construction workers

6406531559700. ✔ The problem of displacement of the people in the community

**Question Number : 35 Question Id : 640653469275 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

The project engineer is giving her opinion. Her perspective would be:

**Options :**

6406531559701. ✖ The loss of open spaces for the community

6406531559702. ✖ The air pollution that will follow the construction

6406531559703. ✖ The human cost of the development project

6406531559704. ✔ The primary, secondary, and tertiary benefits of the project for the community

**Question Number : 36 Question Id : 640653469276 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

A family that has been staying there for generations and has a shop in the vicinity is giving their opinion. Their perspective would be:

**Options :**

- 6406531559705. ✖ Fear of displacement
- 6406531559706. ✖ Loss of livelihood
- 6406531559707. ✖ Neither Fear of displacement nor Loss of livelihood
- 6406531559708. ✔ Both Fear of displacement and Loss of livelihood

**Question Number : 37 Question Id : 640653469277 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

An unemployed young man from the area is stating his opinion. His perspective would be:

**Options :**

- 6406531559709. ✖ Hope for employment opportunities with the project
- 6406531559710. ✖ Concern for displacement of his friends and family
- 6406531559711. ✔ Both Hope for employment opportunities with the project and Concern for displacement of his friends and family
- 6406531559712. ✖ Neither Hope for employment opportunities with the project nor Concern for displacement of his friends and family

|                              |             |
|------------------------------|-------------|
| Sub-Section Number :         | 5           |
| Sub-Section Id :             | 64065367470 |
| Question Shuffling Allowed : | Yes         |
| Is Section Default? :        | null        |

**Question Number : 38 Question Id : 640653469284 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Select true or false for the following statement.

*'Bring out'* is a separable phrasal verb.

**Options :**

6406531559738. ✓ True

6406531559739. ✗ False

**Question Number : 39 Question Id : 640653469285 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Fill in the blank with the appropriate option.

The meaning of the phrasal verb *'rule out'* is \_\_\_\_\_.

**Options :**

6406531559740. ✓ To state that something is impossible

6406531559741. ✗ To faint or lose consciousness

6406531559742. ✗ To extinguish a fire

**Question Number : 40 Question Id : 640653469286 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Select true or false for the following statement.

The idiom '*to hear on the grapevine*' means '*to hear a rumour or an unconfirmed story.*'

**Options :**

6406531559743. ✓ True

6406531559744. ✗ False

**Question Number : 41 Question Id : 640653469287 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the correct option.

\_\_\_\_\_ you play with us?

**Options :**

6406531559745. ✗ Shall

6406531559746. ✓ Will

6406531559747. ✗ May

6406531559748. ✗ All of these

**Question Number : 42 Question Id : 640653469288 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the correct option.

\_\_\_\_\_ we leave now or do you want to wait?

**Options :**

6406531559749. ✗ Will

6406531559750. ✓ Shall

**Question Number : 43 Question Id : 640653469289 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Fill in the blank with an appropriate article:

Come for \_\_\_\_\_ interview tomorrow.

**Options :**

6406531559751. ✖ A

6406531559752. ✔ An

6406531559753. ✖ The

6406531559754. ✖ No article

**Question Number : 44 Question Id : 640653469290 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which one of the following words is a proper noun?

**Options :**

6406531559755. ✔ Jimmy

6406531559756. ✖ Police

6406531559757. ✖ Government

6406531559758. ✖ Hardship

**Question Number : 45 Question Id : 640653469291 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Select the correct pronoun.

I took the bag from \_\_\_?

**Options :**

6406531559759. ✓ Her

6406531559760. ✗ They

**Question Number : 46 Question Id : 640653469292 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Complete the sentence by choosing the correct form of the verb given in brackets.

She was \_\_\_\_\_(bite) by a shark.

**Options :**

6406531559761. ✗ Bite

6406531559762. ✗ Biting

6406531559763. ✗ Bit

6406531559764. ✓ Bitten

**Question Number : 47 Question Id : 640653469293 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction**

**Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which one of the following words is a preposition?

**Options :**

6406531559765. ✗ Ooh!

6406531559766. ✗ Eww!

6406531559767. ✖ Moreover

6406531559768. ✔ Beyond

**Question Number : 48 Question Id : 640653469294 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the appropriate option.

You can \_\_\_\_\_ phone numbers in a directory.

**Options :**

6406531559769. ✔ Look up

6406531559770. ✖ Look after

6406531559771. ✖ Look up to

6406531559772. ✖ Look in

**Question Number : 49 Question Id : 640653469295 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Indian English is characterised by the use of retroflex sounds. This statement is

**Options :**

6406531559773. ✔ TRUE

6406531559774. ✖ FALSE

**Question Number : 50 Question Id : 640653469296 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

There are \_\_\_ vowel sounds in English language.

**Options :**

6406531559775. ✖ 5

6406531559776. ✖ 10

6406531559777. ✖ 15

6406531559778. ✔ 20

**Question Number : 51 Question Id : 640653469297 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which among the following has /ei/ sound.

**Options :**

6406531559779. ✖ Kate

6406531559780. ✖ Cat

6406531559781. ✖ Weight

6406531559782. ✔ Both Kate and Weight

**Question Number : 52 Question Id : 640653469298 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which among the following words has a diphthong?

**Options :**

6406531559783. ✖ Tear

6406531559784. ✖ Share

6406531559785. ✖ Litre



6406531559786. ✔ Both Tear and Share

Question Number : 53 Question Id : 640653469299 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

The word '*thermometer*' begins with a \_\_\_\_ sound.

Options :

6406531559787. ✖ Palatal

6406531559788. ✖ Velar

6406531559789. ✔ Dental

6406531559790. ✖ Labial

CT

|                                                              |             |
|--------------------------------------------------------------|-------------|
| Section Id :                                                 | 64065330301 |
| Section Number :                                             | 3           |
| Section type :                                               | Online      |
| Mandatory or Optional :                                      | Mandatory   |
| Number of Questions :                                        | 15          |
| Number of Questions to be attempted :                        | 15          |
| Section Marks :                                              | 50          |
| Display Number Panel :                                       | Yes         |
| Group All Questions :                                        | No          |
| Enable Mark as Answered Mark for Review and Clear Response : | Yes         |
| Maximum Instruction Time :                                   | 0           |
| Sub-Section Number :                                         | 1           |
| Sub-Section Id :                                             | 64065367471 |

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 54 Question Id : 640653469300 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "QUALIFIER LEVEL : COMPUTATIONAL THINKING"  
MANDATORILY YOU HAVE TO ATTEND ALL THE SECTIONS

Options :

6406531559791. ✓ YES

6406531559792. ✗ NO

Question Number : 55 Question Id : 640653469301 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

## Scores

| SeqNo | Name      | Gender | DateOfBirth | TownCity | Mathematics | Physics | Chemistry | Total |
|-------|-----------|--------|-------------|----------|-------------|---------|-----------|-------|
| 0     | Bhuvanesh | M      | 7 Nov       | Erode    | 68          | 64      | 78        | 210   |
| ■ ■ ■ |           |        |             |          |             |         |           |       |
| 29    | Naveen    | M      | 13 Oct      | Vellore  | 72          | 66      | 81        | 219   |

## Words

| SeqNo | Word  | PartOfSpeech | LetterCount |
|-------|-------|--------------|-------------|
| 0     | It    | Pronoun      | 2           |
| ■ ■ ■ |       |              |             |
| 64    | cane. | Noun         | 4           |

## Library

| SeqNo | Name           | Author  | Genre      | Language | Pages | Publisher      | Year |
|-------|----------------|---------|------------|----------|-------|----------------|------|
| 0     | Igniting Minds | Kalam   | Nonfiction | English  | 178   | Penguin        | 2002 |
| ■ ■ ■ |                |         |            |          |       |                |      |
| 29    | Malgudi Days   | Narayan | Fiction    | English  | 150   | Indian Thought | 1943 |

| Olympics |                   |        |             |              |      |               |        |
|----------|-------------------|--------|-------------|--------------|------|---------------|--------|
| SeqNo    | Name              | Gender | Nationality | Host country | Year | Sport         | Medal  |
| 0        | Karnam Malleswari | F      | Indian      | Australia    | 2000 | Weightlifting | Bronze |
| - - -    |                   |        |             |              |      |               |        |
| 49       | Michael Phelps    | M      | American    | China        | 2008 | Swimming      | Gold   |

## Three sample cards out of 30 for Shopping Bills dataset

Item List

| SV Stores |                  | Srivatsan 1 |       |      |
|-----------|------------------|-------------|-------|------|
| Item      | Category         | Qty         | Price | Cost |
| Carrots   | Vegetables/Food  | 1.5         | 50    | 75   |
| Soap      | Toiletries       | 4           | 32    | 128  |
| Tomatoes  | Vegetables/Food  | 2           | 40    | 80   |
| Bananas   | Vegetables/Food  | 8           | 8     | 64   |
| Socks     | Footwear/Apparel | 3           | 56    | 168  |
| Curd      | Dairy/Food       | 0.5         | 32    | 16   |
| Milk      | Dairy/Food       | 1.5         | 24    | 36   |
|           |                  |             |       | 567  |

| Sun General    |             | Vignesh 14 |       |      |
|----------------|-------------|------------|-------|------|
| Item           | Category    | Qty        | Price | Cost |
| Phone Charger  | Utilities   | 1          | 230   | 230  |
| Razor Blades   | Grooming    | 1          | 12    | 12   |
| Razor          | Grooming    | 1          | 45    | 45   |
| Shaving Lotion | Grooming    | 0.8        | 180   | 144  |
| Earphones      | Electronics | 1          | 210   | 210  |
| Pencils        | Stationery  | 3          | 5     | 15   |
|                |             |            |       | 656  |

| Big Bazaar    |                 | Sudeep 2 |       |      |
|---------------|-----------------|----------|-------|------|
| Item          | Category        | Qty      | Price | Cost |
| Baked Beans   | Canned/Food     | 1        | 125   | 125  |
| Chicken Wings | Meat/Food       | 0.5      | 600   | 300  |
| Cocoa powder  | Canned/Food     | 1        | 160   | 160  |
| Capsicum      | Vegetables/Food | 0.8      | 180   | 144  |
| Tie           | Apparel         | 2        | 390   | 780  |
| Clips         | Household       | 0.5      | 32    | 16   |
|               |                 |          |       | 1525 |

Options :

6406531559793. ✓ Useful Data has been mentioned above.

6406531559794. ✖ This data attachment is just for a reference & not for an evaluation.

Sub-Section Number :

2

Sub-Section Id :

64065367472

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 56 Question Id : 640653469302 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Let X and Y be two rows in the "Words" dataset. Select the most appropriate datatype for each

item/expression in the left column.

| Field               | Data Type  |
|---------------------|------------|
| a. X.Word == Y.Word | 1. String  |
| b. X.Word == "a"    | 2. Invalid |
| c. X.PartOfSpeech   | 3. Boolean |
| d. X.LetterCount    | 4. Integer |

**Options :**

6406531559795. ✔ a - (3), b - (3), c - (1), d - (4)

6406531559796. ✖ a - (1), b - (3), c - (4), d - (3)

6406531559797. ✖ a - (3), b - (2), c - (1), d - (4)

6406531559798. ✖ a - (1), b - (2), c - (3), d - (4)

**Question Number : 57 Question Id : 640653469303 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

The following pseudocode is executed using the "Words" dataset. What will **count** represent at the end of the execution?

```

1  count = 0
2  while(Table 1 has more rows){
3      flag1 = False, flag2 = False
4      Read the first row X in Table 1
5      if(X.PartOfSpeech == "Noun"){
6          flag1 = True
7      }
8      if(X.LetterCount >= 4){
9          flag2 = True
10     }
11     if(flag1 or flag2){
12         count = count + 1
13     }
14     Move X to Table 2
15 }

```

### Options :

6406531559799. ✖ Number of words which are nouns and have at least four letters

6406531559800. ✖ Number of words which are either nouns or have at least four letters but not both

6406531559801. ✔ Number of words which are either nouns or have at least four letters or both

6406531559802. ✖ Number of words which are not nouns and have at most three letters

**Sub-Section Number :** 3

**Sub-Section Id :** 64065367473

**Question Shuffling Allowed :** Yes

**Is Section Default? :** null

**Question Number : 58 Question Id : 640653469304 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Multiple Choice Question

The following pseudocode is executed using the "Scores" dataset. What will **count2** represent at the end of the execution?

```

1  count1 = 0, count2 = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      if(X.Gender == 'M' or X.Mathematics > X.Physics){
5          count1 = count1 + 1
6      }
7      else{
8          count2 = count2 + 1
9      }
10     Move X to Table 2
11 }

```

### Options :

6406531559803. ✖ Number of male students whose Physics marks are greater than Mathematics marks

6406531559804. ✖ Number of male students whose Physics marks are greater than or equal to Mathematics marks

6406531559805. ✔ Number of female students whose Physics marks are greater than or equal to Mathematics marks

6406531559806. ✖ Number of female students whose Physics marks are less than or equal to Mathematics marks

**Question Number : 59 Question Id : 640653469305 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Multiple Choice Question

The following pseudocode is executed using the "Library" dataset. Procedure **biGenre(A)** returns True if the author **A** has written equal number of fictional and non-fictional books. Choose the correct code fragment to complete the pseudocode. Assume that the dataset has only two possible genres.

```

1 Procedure biGenre(A)
2     count = 0
3     while(Table 1 has more rows){
4         Read the first row X in Table 1
5         Move X to Table 2
6         *****
7         * Fill the code *
8         *****
9     }
10    if(count == 0){
11        return(True)
12    }
13    return(False)
14 End biGenre

```

Options :

```

1  if(X.Author == A and X.Genre == "Fiction"){
2      count = count + 1
3  }
4  else{
5      if(X.Author == A){
6          count = count - 1
7      }
8  }

```

6406531559807. ✓

```

1  if(X.Author == A and X.Genre == "Fiction"){
2      count = count + 1
3  }
4  else{
5      count = count - 1
6  }

```

6406531559808. ✖

```

1  if(X.Author == A){
2      count = count + 1
3  }
4  if(X.Genre == "Fiction"){
5      count = count + 1
6  }
7  else{
8      count = count - 1
9  }

```

6406531559809. ✖



```

1  if(X.Author == A and X.Genre == "Fiction"){
2      count = count + 1
3  }
4  else{
5      if(X.Genre != "Fiction"){
6          count = count - 1
7      }
8  }

```

6406531559810. ✖

**Question Number : 60 Question Id : 640653469306 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Multiple Choice Question

The following pseudocode is executed using the "Olympics" dataset. What will **count** represent at the end of the execution? Assume that every player has a distinct name.

```

1  count = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      Move X to Table 2
5      flag = False
6      while(Table 1 has more rows){
7          Read the first row Y in Table 1
8          if(X.Name == Y.Name){
9              if((X.Sport != Y.Sport) and (X.Medal == Y.Medal)){
10                 flag = True
11             }
12             Move Y to Table 2
13         }
14         else{
15             Move Y to Table 3
16         }
17     }
18     if(flag){
19         count = count + 1
20     }
21     Move all rows from Table 3 to Table 1
22 }

```

### Options :

6406531559811. ✖ Number of players who have won different medals in the same sport
6406531559812. ✔ Number of players who have won the same medal in different sports
6406531559813. ✖ Number of pairs of players who have won different medals in the same sport
6406531559814. ✖ Number of pairs of players who have won the same medal in different sports

**Sub-Section Number :** 4

**Sub-Section Id :** 64065367474

**Question Shuffling Allowed :** Yes

**Is Section Default? :** null

**Question Number : 61 Question Id : 640653469307 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4 Selectable Option : 0**

Question Label : Multiple Select Question

The following pseudocode is executed using the "Library" dataset. Assume there are **n** books with distinct names that are written by **m** authors and each book is written by only one author. Assume that every author has a distinct name.

```
1  count1 = 0, count2 = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      Move X to Table 2
5      count1 = count1 + 1
6      while(Table 1 has more rows){
7          Read the first row Y in Table 1
8          if(X.Author == Y.Author){
9              count2 = count2 + 1
10             Move Y to Table 2
11         }
12         else{
13             Move Y to Table 3
14         }
15     }
16     Move all rows from Table 3 to Table 1
17 }
```

At the end of the execution of the above pseudocode, choose the correct option(s). It is a Multiple Select Question (MSQ).

**Options :**

6406531559815. ✖ count1 = n

6406531559816. ✔ count1 = m

6406531559817. ✖ count2 = m - n

6406531559818. ✔ count2 = n-m

**Question Number : 62 Question Id : 640653469308 Question Type : MSQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4 Selectable Option : 0**

Question Label : Multiple Select Question

The following pseudocode is executed using the "Scores" dataset. At the end of the execution, **A** captures the number of students who are either male from Bengaluru or have scored lower marks in Physics than the average Physics marks. Assume that the variable **Avg** holds the value of the average Physics marks. But the pseudocode may have mistakes. Identify all such mistakes (if any). Assume that all statements not listed in the options below are free of errors.

It is a Multiple Select Question (MSQ).

```

1  A = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      C = False, D = False
5      if(X.Gender == 'M' or X.TownCity == "Bengaluru"){
6          C = True
7      }
8      if(X.Physics < Avg){
9          D = True
10     }
11     if(C or D){
12         A = A + 1
13     }
14     Move X to Table 2
15 }

```

Options :

6406531559819. ✖ Line 1: Incorrect initialization of **A**

6406531559820. ✔ Line 5: Condition to update **C** is incorrect

6406531559821. ✖ Line 11: Condition to update **A** is incorrect

6406531559822. ✖ No error in the code

Sub-Section Number :

5

Sub-Section Id :

64065367475

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 63 Question Id : 640653469309 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

The following pseudocode is executed using the "Scores" dataset. At the end of the execution, **count** captures the number of pairs of students who are of the same gender or are from the same city but not both. Choose the correct code fragment to complete the pseudocode.

It is a Multiple Select Question (MSQ).

```

1  count = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      Move X to Table 2
5      while(Table 1 has more rows){
6          Read the first row Y in Table 1
7          Move Y to Table 3
8          count = count + findPair(X, Y)
9      }
10     Move all rows from Table 3 to Table 1
11 }
12 Procedure findPair(X, Y)
13     *****
14     ***** Fill the code *****
15     *****
16 End findPair

```

Options :

```

1  A = 0, B = 0
2  if(X.Gender == Y.Gender or X.TownCity == Y.TownCity){
3      A = A + 1
4  }
5  if(X.Gender == Y.Gender and X.TownCity == Y.TownCity){
6      B = B + 1
7  }
8  return(A-B)

```

6406531559823. ✓

```

1  A = 0, B = 0
2  if(X.Gender == Y.Gender and X.TownCity == Y.TownCity){
3      A = A + 1
4  }
5  if(X.Gender == Y.Gender or X.TownCity == Y.TownCity){
6      B = B + 1
7  }
8  return(A-B)

```

6406531559824. ✗

6406531559825. ✓

```
1 A = False, B = False
2 if(X.Gender == Y.Gender){
3     A = True
4 }
5 if(X.TownCity == Y.TownCity){
6     B = True
7 }
8 if((A and not B) or (not A and B)){
9     return(1)
10 }
11 return(0)
```

```
1 A = False, B = False
2 if(X.Gender == Y.Gender){
3     A = True
4 }
5 if(X.TownCity == Y.TownCity){
6     B = True
7 }
8 if((A or not B) and (not A or B)){
9     return(1)
10 }
11 return(0)
```

6406531559826. ✖

**Question Number : 64 Question Id : 640653469310 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 5 Selectable Option : 0**

Question Label : Multiple Select Question



Two words are said to be conjugate if they fulfill following conditions:

- They are different words
- Number of vowels are same in both words
- Number of consonants are different in both words

For a row **X** in the "Words" dataset, assume that **vCount(X)** return the number of vowels in **X.Word**. At the end of the execution, **count** stores the number of conjugate pairs. Choose the correct code fragment(s) to complete the pseudocode. It is a Multiple Select Question (MSQ).

```
1  count = 0
2  while(Table 1 has more rows){
3      Read the first row X in Table 1
4      Move X to Table 2
5      while(Table 1 has more rows){
6          Read the first row Y in Table 1
7          *****
8          * Fill the code *
9          *****
10         Move Y to Table 3
11     }
12     Move all rows from Table 3 to Table 1
13 }
```

Options :

```
1  if(X.word == Y.word){
2      if(X.LetterCount == Y.LetterCount){
3          if(vCount(X) == vCount(Y)){
4              count = count + 1
5          }
6      }
7  }
```

6406531559827. ✖

```
1  if(X.word != Y.word){
2      if(X.LetterCount != Y.LetterCount){
3          if(vCount(X) == vCount(Y)){
4              count = count + 1
5          }
6      }
7  }
```

6406531559828. ✔

6406531559829. ✖

```

1  if(X.Word != Y.Word){
2      if(vCount(X) == vCount(Y)){
3          if(X.LetterCount - vCount(X) == Y.LetterCount - vCount(Y)){
4              count = count + 1
5          }
6      }
7  }

```

```

1  if(X.Word != Y.Word){
2      if(vCount(X) == vCount(Y)){
3          if(X.LetterCount - vCount(X) != Y.LetterCount - vCount(Y)){
4              count = count + 1
5          }
6      }
7  }

```

6406531559830. ✓

**Sub-Section Number :**

6

**Sub-Section Id :**

64065367476

**Question Shuffling Allowed :**

Yes

**Is Section Default? :**

null

**Question Number : 65 Question Id : 640653469311 Question Type : SA Calculator : None**

**Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4**

Question Label : Short Answer Question

The following pseudocode is executed using a dataset similar to the "Words" dataset, based on the following paragraph.

"The product which I selected yesterday had a design on the left pocket. The color is also different. I clearly remember that the color which I had chosen was slightly dark. I can not believe that the shopkeeper has sent me a different product."



```

1  count1 = 0, count2 = 0, count = 0
2  flag = True
3  while(Table 1 has more rows){
4      Read the first row x in Table 1
5      count1 = count1 + 1
6      Move x to Table 2
7      if(flag){
8          if(x.word == "the"){
9              count2 = count2 + 1
10         }
11     }
12     flag = False
13     if(x.word ends with full stop){
14         flag = True
15     }
16 }
17 count = (count2 * 100)/count1

```

What would be the value of **count** at the end of the execution of the above pseudocode? Assume that upper case and lower case are ignored during comparison of letters.

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

50

**Question Number :** 66 **Question Id :** 640653469312 **Question Type :** SA **Calculator :** None

**Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0

**Correct Marks :** 4

**Question Label :** Short Answer Question

What would be the value of **result** at the end of the execution of the following pseudocode if the value of **n** is 7?

```
1  i = 1
2  result1 = 1, result2 = 1, result = 1
3  while(i < n){
4      result1 = result1*(i+1)
5      result2 = result2*i*(i + 1)
6      i = i + 2
7  }
8
9  result = result2/result1
```

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

15

**Question Number :** 67 **Question Id :** 640653469313 **Question Type :** SA **Calculator :** None

**Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0

**Correct Marks :** 4

**Question Label :** Short Answer Question

The following pseudocode is executed using the “Words” dataset.

```

1  Procedure dosomething(Table T1, Table T2)
2      count = 0
3      while(Table T1 has more rows){
4          Read the first row Y from Table T1
5          Read the first row Z from Table T2
6          if(Y.word == Z.word){
7              count = count + 1
8          }
9          else{
10             return(count)
11         }
12         Move the row Y to Table T1
13         Move the row Z to Table T2
14     }
15     return(count)
16 End dosomething

```

Let there be two datasets stored in Table 1 and Table 2 corresponding to the following sentences

Table 1:

"if you are honest with your work, then the money is just a compliment"

Table 2:

"if you are honest with your work, then the progress is just a compliment"

What will **doSomething**(Table 1, Table 2) return? Assume that upper case and lower case are ignored during comparison of words.

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

9

**Sub-Section Number :** 7

**Sub-Section Id :** 64065367477

**Question Shuffling Allowed :** No

Is Section Default? :

null

**Question Id : 640653469314 Question Type : COMPREHENSION Sub Question Shuffling**  
**Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix**  
**Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**  
**Question Numbers : (68 to 69)**

Question Label : Comprehension

The following pseudocode is executed using the "Scores" dataset. Let variables F, M, F250, and M250 represent the followings:

- F = Number of female students in the dataset
- M = Number of male students in the dataset
- F250 = Number of female students with total marks greater than 250
- M250 = Number of male students with total marks greater than 250

```
1  A = 0, B = 0
2  while(Table 1 has more rows){
3      Read the first row X from Table 1
4      flag = False
5      if(X.Gender == 'M'){
6          if(X.Total > 250){
7              flag = True
8          }
9      }
10     if(not flag){
11         A = A + 1
12         if(X.Total > 250){
13             B = B + 1
14         }
15     }
16     Move X to Table 2
17 }
```

Based on the above data, answer the given subquestions.

**Sub questions**

**Question Number : 68 Question Id : 640653469315 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Multiple Choice Question

What will be the value of **A** at the end of the execution?

**Options :**

6406531559834. ✖  $F + F250$

6406531559835. ✔  $F + (M - M250)$

6406531559836. ✖  $M + (F - F250)$

6406531559837. ✖  $M + F250$

**Question Number : 69 Question Id : 640653469316 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4**

Question Label : Multiple Choice Question

What will be the value of **B** at the end of the execution?

**Options :**

6406531559838. ✖  $M$

6406531559839. ✔  $F250$

6406531559840. ✖  $M250$

6406531559841. ✖  $M + (F - F250)$

## Stats-1

|                         |             |
|-------------------------|-------------|
| Section Id :            | 64065330302 |
| Section Number :        | 4           |
| Section type :          | Online      |
| Mandatory or Optional : | Mandatory   |
| Number of Questions :   | 11          |

|                                                              |             |
|--------------------------------------------------------------|-------------|
| Number of Questions to be attempted :                        | 11          |
| Section Marks :                                              | 40          |
| Display Number Panel :                                       | Yes         |
| Group All Questions :                                        | No          |
| Enable Mark as Answered Mark for Review and Clear Response : | Yes         |
| Maximum Instruction Time :                                   | 0           |
| Sub-Section Number :                                         | 1           |
| Sub-Section Id :                                             | 64065367478 |
| Question Shuffling Allowed :                                 | No          |
| Is Section Default? :                                        | null        |

Question Number : 70 Question Id : 640653469317 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0  
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "QUALIFIER LEVEL : STATISTICS FOR DATA SCIENCE I"

MANDATORILY YOU HAVE TO ATTEND ALL THE SECTIONS

Options :

6406531559842. ✓ YES

6406531559843. ✗ NO

|                              |             |
|------------------------------|-------------|
| Sub-Section Number :         | 2           |
| Sub-Section Id :             | 64065367479 |
| Question Shuffling Allowed : | No          |
| Is Section Default? :        | null        |

Question Id : 640653469318 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

**Question Numbers : (71 to 72)**

Question Label : Comprehension

Figure Q.1 represents the placement report of an engineering college for the academic year 2021-22.

Placement report

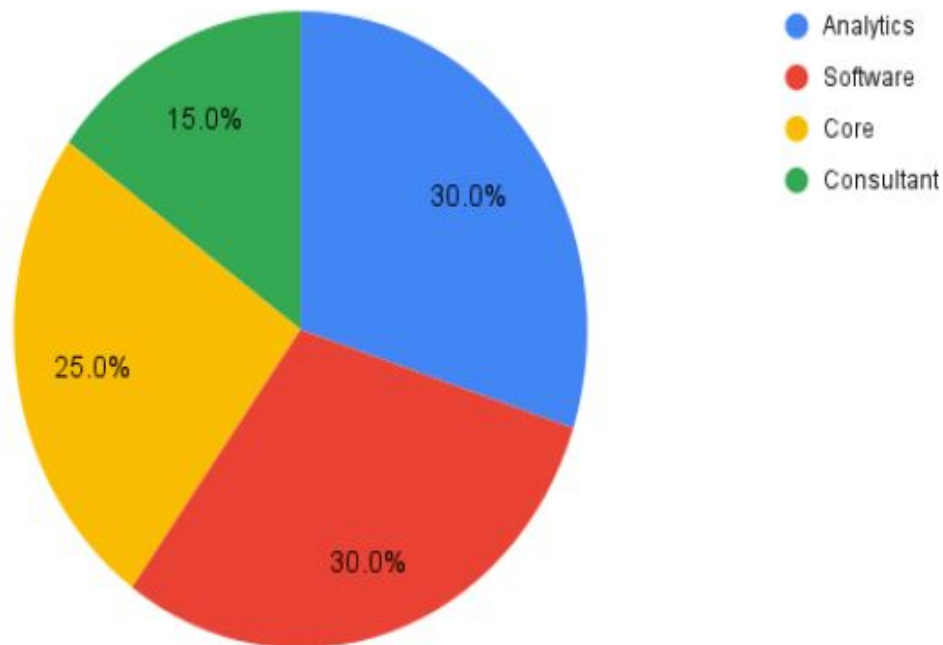


Figure Q.1: Placement report of an engineering college

Based on the information, answer the given subquestions.

**Sub questions**

**Question Number : 71 Question Id : 640653469319 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Which of the following category **does not** represent the mode?

**Options :**

6406531559844. ✖ Analytics

6406531559845. ✖ Software

6406531559846. ✓ Core

6406531559847. ✖ Insufficient information

**Question Number : 72 Question Id : 640653469320 Question Type : SA Calculator : None**

**Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Short Answer Question

If 1000 students are placed during the placement season of 2021-22, then find the number of students placed in core companies?

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Equal**

**Text Areas : PlainText**

**Possible Answers :**

250

**Sub-Section Number :** 3

**Sub-Section Id :** 64065367480

**Question Shuffling Allowed :** Yes

**Is Section Default? :** null

**Question Number : 73 Question Id : 640653469321 Question Type : MSQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3 Selectable Option : 0**

Question Label : Multiple Select Question

Which of the following statements is/are true?

**Options :**

6406531559849. ✓ A sample is the subset of a population.

6406531559850. ✓ Numerical variables can have all the properties of ordinal and nominal scales



of measurement.

6406531559851. ✖ Categorical variable can have numerical properties.

6406531559852. ✔ Correlation coefficient measures a linear association between two numerical variables.

**Question Number : 74 Question Id : 640653469327 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3 Selectable Option : 0**

Question Label : Multiple Select Question

Which of the following represent(s) cross sectional data?

**Options :**

6406531559866. ✔ Census of tigers estimated in four different countries in 2021.

6406531559867. ✖ Yearly irrigation report released by a government over the period of ten years.

6406531559868. ✔ Revenue generated by a company in four different states in 2022.

6406531559869. ✖ Monthly expenditure of a family in 2020.

**Question Number : 75 Question Id : 640653469328 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3 Selectable Option : 0**

Question Label : Multiple Select Question

Choose the correct statement/s ?

**Options :**

6406531559870. ✔ A nominal scale has the property of labelling the categories and it does not involve the ranking of data.

6406531559871. ✔ An ordinal scale has all the properties of nominal scale and it involves the ranking of data.

6406531559872. ✖ An Interval scale has all the properties of ordinal scale and it satisfies the absolute zero property.

6406531559873. ✖ A ratio scale has all the properties of interval scale and it does not satisfy the

absolute zero property.

Sub-Section Number : 4  
Sub-Section Id : 64065367481  
Question Shuffling Allowed : Yes  
Is Section Default? : null

Question Number : 76 Question Id : 640653469322 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Selectable Option : 0

Question Label : Multiple Select Question

Suppose the number of cars sold by a car dealer in five months are 1, 12, 8, 9, 20. Which of the following is/are outliers?

Options :

- 6406531559853. ✓ 1
- 6406531559854. ✓ 20
- 6406531559855. ✗ 12
- 6406531559856. ✗ None of these

Question Number : 77 Question Id : 640653469323 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Selectable Option : 0

Question Label : Multiple Select Question

An analyst did a survey to know whether literacy rate affects voting rate among adults in villages. He collected the data from a village and the results are tabulated in Table Q.1.

| literacy | Participated in voting |    |
|----------|------------------------|----|
|          | Yes                    | No |
| Yes      | 30                     | 10 |
| No       | 40                     | 15 |

Table Q.1

Choose the correct option/s?

**Options :**

6406531559857. ✓ Of all the literate adults, 25% do not participate in voting.

6406531559858. ✓ Of all the illiterate adults, 72.73% participate in voting.

6406531559859. ✓ If all row relative frequencies are similar within each column then, it implies that all column relative frequencies will also be similar within each row.

6406531559860. ✖ If all column relative frequencies are similar within each row then, it does not imply that all row relative frequencies will also be similar within each column.

6406531559861. ✓ There is no association between literacy and voting participation.

6406531559862. ✖ There is an association between literacy and voting participation.

**Sub-Section Number :**

5

**Sub-Section Id :**

64065367482

**Question Shuffling Allowed :**

No

**Is Section Default? :**

null

**Question Id : 640653469324 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Question Numbers : (78 to 79)**

Question Label : Comprehension

An analyst want to analyse the salary of employees in different organizations in a city. To analyse this, he has selected an organization and the data of salaries is tabulated as shown in Table Q.2.

| Employee name | Post              | Salary (in thousands rupees) |
|---------------|-------------------|------------------------------|
| Sagar         | Assistant manager | 70                           |
| Sanjana       | Analyst           | 45                           |
| Rohit         | Junior developer  | 50                           |
| Jenny         | Senior developer  | 60                           |
| Ross          | Consultant        | 50                           |

Table Q.2

Based on the information, answer the given subquestions.

### Sub questions

**Question Number : 78 Question Id : 640653469325 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Based on the data collected from an organisation, analyst made a statement that the average salary of an employee is 50,000 rupees in different organizations in the city. The given statement of analyst is based on which kind of statistical analysis ?

**Options :**

6406531559863. ✖ Descriptive statistics

6406531559864. ✔ Inferential statistics

**Question Number : 79 Question Id : 640653469326 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 4**

Question Label : Short Answer Question

What is the sample standard deviation of salary (in thousand rupees)?

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Range**

**Text Areas : PlainText**

**Possible Answers :**

9.7 to 10.3

**Question Id : 640653469329 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Question Numbers : (80 to 81)**

Question Label : Comprehension

The heights (in cm) of ten athletes are represented by a stem and leaf plot in Figure Q.2.

| Stem | Leaf    |
|------|---------|
| 16   | 2 2 8 9 |
| 17   | 5 5 5 7 |
| 18   | 0 4     |

Here 16 | 4 represents 164 cm.

Figure Q.2

Based on the information, answer the given subquestions.

**Sub questions****Question Number : 80 Question Id : 640653469330 Question Type : SA Calculator : None****Response Time : N.A Think Time : N.A Minimum Instruction Time : 0****Correct Marks : 3**

Question Label : Short Answer Question

What is the median height(in cm) of an athlete?

**Response Type : Numeric****Evaluation Required For SA : Yes****Show Word Count : Yes****Answers Type : Equal****Text Areas : PlainText****Possible Answers :**

175

**Question Number : 81 Question Id : 640653469331 Question Type : SA Calculator : None****Response Time : N.A Think Time : N.A Minimum Instruction Time : 0****Correct Marks : 3**

Question Label : Short Answer Question

What is the mode of the heights (in cm) of athletes?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

175

Sub-Section Number : 6

Sub-Section Id : 64065367483

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 82 Question Id : 640653469332 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

The mean and sample variance of the marks (out of 50) scored by the students in the Statistics course are 40 and 25 respectively. If 5 is added to the marks of each student and then multiplied by 2. Find the mean of the modified marks.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

90

Sub-Section Number : 7

Sub-Section Id : 64065367484

**Question Shuffling Allowed :**

Yes

**Is Section Default? :**

null

**Question Number : 83 Question Id : 640653469333 Question Type : MCQ Is Question**

**Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

**Correct Marks : 3**

Question Label : Multiple Choice Question

Let  $X$  and  $Y$  represent the total marks of Mayur and Deepak in 5 subjects respectively. If the marks scored by Deepak is thrice the marks scored by Mayur in each subject, then which of the following statement is correct?

**Options :**

6406531559877. ✖ There is no correlation between  $X$  and  $Y$ .

6406531559878. ✖ There is a weak positive correlation between  $X$  and  $Y$ .

6406531559879. ✔ There is a strong positive correlation between  $X$  and  $Y$ .

6406531559880. ✖ Insufficient information.